

Advanced (Habanero)

Open Ended Questions (FRQ)

Q9. Tom has been saving money for a new bike and has 120 in his piggy bank. He wants to buy a bike that costs 180. Tom's parents have agreed to give him an allowance of 5 per week for doing chores, and his grandmother has agreed to give him 10 per week. How many weeks will it take Tom to have enough money to buy the bike? Show all work and explain your answer.

Q10. A book store sells a combination of fiction and non-fiction books. The store sells x fiction books at

10

each and y non-fiction books at

15

each. The total number of books sold is 240, and the total revenue from book sales is

2160

. Write a system of equations to represent this situation, and solve for x and y . Show all work.

Chef's Tips

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- Tip 1: Encourage students to read the problem carefully and identify the key elements.
- Tip 2: Remind students to use the given information to write equations.
- Tip 3: Emphasize the importance of checking the reasonableness of the solution.